8.8 Constant Temperature and Pressure Molecular Dynamics

The isobaric, isothermal or NPT ensemble models a system in contact with a heat bath (temperature T) and a "pressure reservoir" maintained at pressure P. The pressure is defined by the formula

$$\mathrm{P} = -\left[\frac{\partial \mathrm{E}}{\partial \mathrm{V}}\right]_{\mathrm{S,N}}$$

where E is the energy, V is volume, S is entropy. The notation $[\ldots]_{x,y}$ means that the indicated partial derivatives should be taken with respect to constant values of the indicated variables.

An isothermal-isobaric system exchanges energy with the bath in such a way as to maintain the constant temperature; at the same time the volume fluctuates to control the pressure. The state of the system can be expressed in terms of particle positions and momenta as well as the volume V, i.e. we think of the probability density as a function of $\boldsymbol{q}, \boldsymbol{p}$ and V.

Let N, V, E be the number of atoms, volume and energy, respectively, of the system of interest, and N_2, V_2, E_2 be the corresponding quantities describing the bath system; we have $\mathrm{N} + \mathrm{N}_2 = \tilde{\mathrm{N}}$, $\mathrm{V} + \mathrm{V}_2 = \tilde{\mathrm{V}}$, $\mathrm{E} + \mathrm{E}_2 = \tilde{\mathrm{E}}$ all constant. We view the entropy as a function of the other variables and expand in a Taylor series in both volume and energy:

$$\mathrm{S}(\mathrm{N}_2, \mathrm{V}_2, \mathrm{E}_2) = \mathrm{S}(\tilde{\mathrm{N}} - \mathrm{N}, \tilde{\mathrm{V}} - \mathrm{V}, \tilde{\mathrm{E}} - \mathrm{E}) \approx \mathrm{S}(\mathrm{N}_2, \tilde{\mathrm{V}}, \tilde{\mathrm{E}}) - \frac{\partial \mathrm{S}}{\partial \mathrm{E}}\mathrm{E} - \frac{\partial \mathrm{S}}{\partial \mathrm{V}}\mathrm{V}.$$

Bear in mind that here E and V are small relative perturbations of the total energy and volume. The quantity $\frac{\partial \mathrm{S}}{\partial \mathrm{E}}$ is the reciprocal temperature.

To calculate $\frac{\partial \mathrm{S}}{\partial \mathrm{V}}$ we may use *Maxwell's thermodynamic relation* [372]:

$$\left[\frac{\partial \mathrm{S}}{\partial \mathrm{E}}\right]_{\mathrm{V,N}} \left[\frac{\partial \mathrm{E}}{\partial \mathrm{V}}\right]_{\mathrm{S,N}} \left[\frac{\partial \mathrm{V}}{\partial \mathrm{S}}\right]_{\mathrm{E,N}} = -1.$$

Since $\left[\frac{\partial \mathrm{S}}{\partial \mathrm{E}}\right]_{\mathrm{V,N}}$ is T^{-1} and $\left[\frac{\partial E}{\partial \mathrm{V}}\right]_{\mathrm{S,N}}$ is $-\mathrm{P}$, we obtain

$$\left[\frac{\partial \mathrm{S}}{\partial \mathrm{V}}\right]_{\mathrm{E,N}} = \frac{1}{\left[\frac{\partial \mathrm{V}}{\partial \mathrm{S}}\right]_{\mathrm{E,N}}} = \mathrm{P}/\mathrm{T}.$$

This results in the formula

$$\mathrm{S}(\mathrm{N}_2, \mathrm{V}_2, \mathrm{E}_2) \approx \text{constant} - \frac{\mathrm{E} + \mathrm{PV}}{\mathrm{T}},$$

which leads to the probability density

$$\rho_{\mathrm{NPT}} = Z_{\mathrm{NPT}}^{-1} e^{-\beta(H+\mathrm{PV})},$$

where H is the Hamiltonian, and $\beta^{-1} = \mathrm{k_B T}$.

The NPT model may be viewed as corresponding to a thermostatted system which incorporates a "piston" controlling the overall volume of the system to be consistent with the target pressure. During simulation, the volume must be free to change; whereas the methods presented so far retain a fixed volume. In NVE or NVT simulation, the volume enters in two ways into the molecular simulation method: (i) as a modification of the potential (allowing for periodic replicas) and (ii) in the mechanism by which atoms are mapped from one boundary to the other of the rectangular domain, simulating motion on a torus. It is not obvious how to generalize these to the NPT setting where V is a dynamical variable.

The method of Andersen [11] ("Andersen's piston") is to control the volume by reparamaterizing the coordinates to lie in a cubic box with unit side. Assuming the system occupies a cubic domain, say, $[-L/2, L/2] \times [-L/2, L/2] \times [-L/2, L/2]$, where $L = V^{1/3}$, we may write, for each three-dimensional atomic coordinate vector $\boldsymbol{q}_i$

$$\boldsymbol{q}_i = V^{1/3} \bar{\boldsymbol{q}}_i, \tag{8.71}$$

where $\bar{\boldsymbol{q}}_i$ is confined to the unit-side cube. Then we have

$$\dot{\boldsymbol{q}}_i = \frac{1}{3} V^{-2/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{1/3} \frac{\mathrm{d}}{\mathrm{d}t} \bar{\boldsymbol{q}}_i.$$

The kinetic energy of the atomic system may be written

$$K = \sum_{i=1}^{N} m_i \frac{\|\dot{\boldsymbol{q}}_i\|^2}{2}.$$

After coordinate transformation this becomes

$$\begin{aligned} \bar{K} &= \sum_{i=1}^{N} \frac{m_i}{2} \left\| \frac{1}{3} V^{-2/3} \dot{V} \bar{\boldsymbol{q}}_i + V^{1/3} \frac{\mathrm{d}}{\mathrm{d}t} \bar{\boldsymbol{q}}_i \right\|^2 \\ &= \sum_{i=1}^{N} \frac{m_i}{2} \left[\frac{1}{9} V^{-4/3} \dot{V}^2 \left\| \bar{\boldsymbol{q}}_i \right\|^2 + \frac{2}{3} V^{-1/3} \dot{V} \bar{\boldsymbol{q}}_i \cdot \dot{\bar{\boldsymbol{q}}}_i + V^{2/3} \left\| \dot{\bar{\boldsymbol{q}}}_i \right\|^2 \right], \end{aligned}$$

thus the momentum vector conjugate to $\bar{\boldsymbol{q}}_i$ becomes

$$\bar{\boldsymbol{p}}_i = \frac{\partial \bar{K}}{\partial \dot{\bar{\boldsymbol{q}}}_i}$$